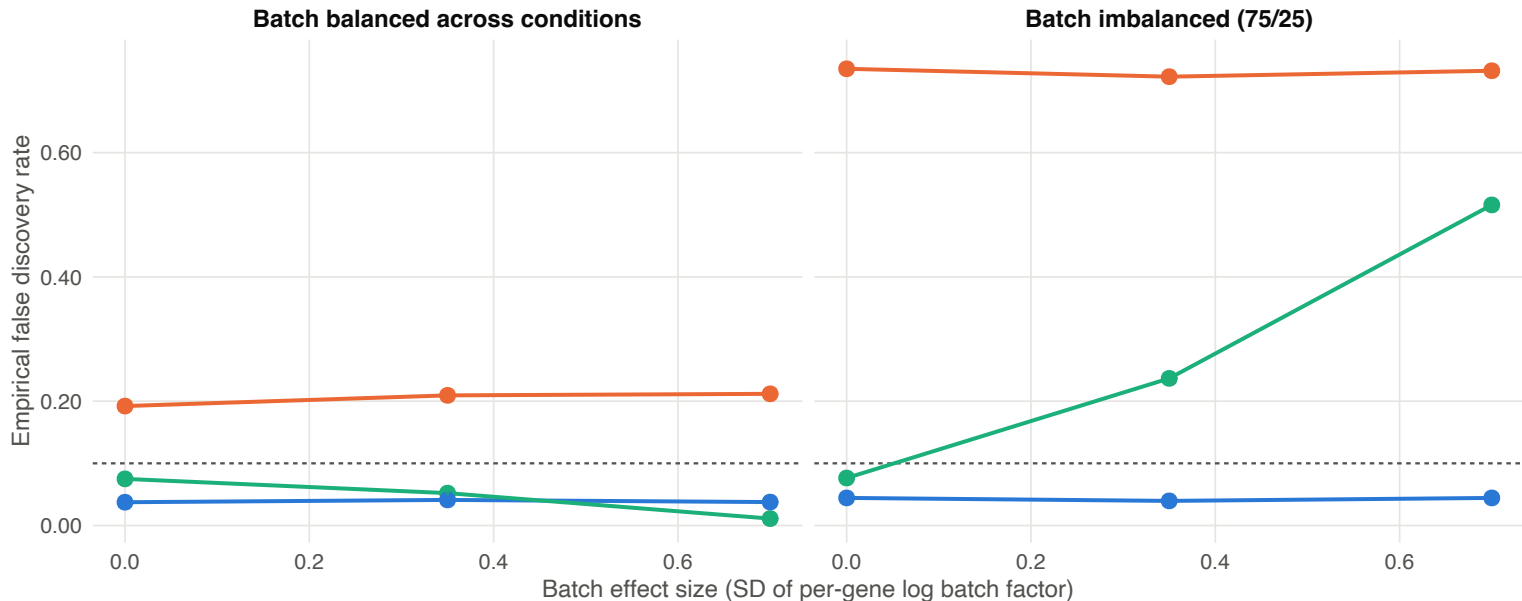


Pre-subtracting batch inflates the false discovery rate

Nominal FDR 0.10 (dashed). n = 5 per group, 50 simulations per point.

CTC pipeline (all decisions) batch pre-subtracted batch not modelled



Additive batch modelling holds at or below nominal; removeBatchEffect does not.